# Aleksei Nemtyrev

Statistical Modelling · Causal Inference · Data Science

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## Profile

Quantitative researcher finishing a PhD in econometrics (2027). My work is about separating signal from noise: I design estimators that cut variance without losing what matters, and evaluate policy rules that could make the economy less volatile. I build them in Python, R and MATLAB and teach Data Science Methods at MSc level. Before my PhD, I ran a retail company, taking it from break-even to profitability within a year.

## Experience

**PhD Candidate** Aug 2023 – present
*Tilburg University, Econometrics & Operations Research* Tilburg, NL

- Develop statistical estimators end to end: theoretical derivation, Monte Carlo evaluation, public reproducible code.
- Run simulations on an HPC cluster (MATLAB Parallel Computing Toolbox): 20M+ bootstrap replications per study.
- Develop macroeconomic (DSGE) models and take them to data, from monetary policy rules to model testing.
- Talks at IAAE 2026, CFE-CMStatistics 2026, GIFT 2026, NESG 2024 & 2026, Econometric Society Winter Meeting 2025; co-authors at NBER-NSF 2026, $(EC)^2$ 2026, Chicago Fed 2025, Purdue 2024, Erasmus 2024.
- Teach *Data Science Methods* to 120 MSc students a year (in R: PCA, ridge/lasso, trees, double ML); designed my own model-selection and PCA assignments. Also teach *Statistics 2* to 300 BSc students a year.

**General Manager** Jun 2016 – Jul 2020
*NEVA Torg (retail)* St. Petersburg, RU

- Ran operations and finances with a team of 10; grew the chain to seven stores (opened three, closed one).
- Redesigned the stores and moved staff pay to a revenue share: revenue per store rose 30% in 2018.
- Renegotiated supplier contracts, lowering the cost of goods by 15% on average and lifting margins.
- Took the company from break-even in 2017 to RUB 2.5m (≈ EUR 35k) after-tax profit in 2018 (∼9% net margin).

## Selected Projects

**Targeted Local Projections** paper code with O. Boldea · working paper, 2026
Optimally combines local projections and VARs, trading a little bias for ∼20% tighter long-horizon confidence intervals.

**FOMC Responses to Credit Spreads** with J. R. Campbell · in progress
Proposes adding credit spreads to the Fed's policy rule; in SVAR simulations, this cuts unemployment volatility by 30%.

**Corrected Local Projections for Structural IRFs** with O. Boldea, A. Cornea-Madeira, A. R. Hall · in progress
Proposes an estimator that strips predictable noise out of local projections; theory and simulations show lower variance.

**DSGE Identification and Testing** with J. R. Campbell · in progress
Novel utility function and model simplification make a macro model estimable and testable directly from data.

## Education

**PhD, Econometrics**, *Tilburg University* 2023 – 2027 (expected)
**Research Master, Economics**, *Tilburg University · CentER Scholarship* 2021 – 2023
**MSc Economics, Monetary Policy & Banking**, *University of Amsterdam · Amsterdam Merit Scholarship* 2020 – 2021
**MA (Hons) Economics**, *University of Glasgow* 2012 – 2016

## Skills

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| **Machine learning** | Bias–variance trade-off, ridge, lasso, cross-validation, PCA, classification, random forests, boosting |
| **Causal inference** | Difference-in-differences, IV, GMM, double ML, structural VARs, local projections, impulse responses |
| **Time series** | Forecasting (VARs, out-of-sample evaluation), ARMA, state-space models, Kalman filter |
| **Statistics** | Bayesian inference (own MCMC sampler in Python), MLE, bootstrap, Monte Carlo simulation |
| **Programming** | Python (NumPy, SciPy, statsmodels, matplotlib), R, MATLAB, SQL, Stata, LaTeX, Git |
| **AI workflows** | Agentic AI (Claude Code, Codex): research codebases, simulation pipelines, document automation |
| **Domain** | Macro-finance, monetary policy, credit spreads, FRED data, DSGE and macro–climate models |
| **Communication** | Data visualisation, research writing, presenting technical results to expert and general audiences |
| **Business** | General management, team leadership, accounting, supplier management, retail operations |
| **Languages** | English (C2), Dutch (B1), Russian (native) |